Homework #5: Nearly free electron model

Due date: Nov 2nd 2025

Compare Kronig-Penney model with the nearly free electron model

1. The Kronig-Penney model allows one to consider both the nearly-free-electron and tight-binding limits. Consider the (repulsive) Dirac-Kronig-Penney model. It gives the following implicit equation for the energy bands:

   $\cos ka = \cos qa + u\frac{\sin qa}{qa}$,

   where $q = \sqrt{2mE}/\hbar$ and $u = \frac{mU_0 a}{\hbar^2} > 0$. Assuming that $u \ll 1$, find the solution of above equation near the Bragg points $k \approx \pm\frac{\pi}{a}, \pm\frac{2\pi}{a}, \ldots$. Compare your result with that of the nearly-free electron model. (Reminder: the $\approx$ sign means that k is close yet not equal to the wavenumber of the Bragg point.)

Nearly free electron model

2. Consider 2D electrons subject to a weak periodic potential

   $U(x,y) = U_0(\cos\frac{2\pi x}{a} + \cos\frac{2\pi y}{a})$.

   (a) Find the energy bands near the edges of the first Brillouin zone. (Hint: the edges of the Brillouin zone are the Bragg planes where the eigenstates are doubly degenerate.) Plot the bands and isoenergetic surfaces.

   (b) Repeat the analysis of part (a) for regions near the corners of the first Brillouin zone, where there are four degenerate eigenstates.